Stroma World — Reality Anchors v1.0

> **What these are:** Observable facts about PDAC stroma biology that the simulation must reproduce qualitatively. If the model violates any of these, something is wired wrong. **What these are not:** Quantitative benchmarks. Exact numbers come later during calibration. These are pass/fail behavioral tests. **How to use them:** Run each anchor as a standalone scenario before committing to full EA runs. Each one has a setup, an expected outcome, and a failure interpretation.

Anchor 1: The Barrier Self-Assembles Without External Instruction

Biological fact

PDAC stroma is not pre-existing. It is *built* by the tumor. Place tumor cells next to quiescent stellate cells and the stroma will form autonomously through paracrine signaling.

Simulation test

Setup: Seed a cluster of tumor cells (canonical PDAC genotype) surrounded by quiescent PSCs (ACTA2=OFF) in a clean domain. No pre-placed ECM. No external signals. No drug. No immune cells. **Run:** Let the simulation evolve for the equivalent of 2–4 weeks simulated time. **Expected outcome:** PSCs nearest the tumor transition to CAFs (ACTA2=ON). ECM density rises concentrically around the tumor, densest at the tumor-stroma interface. A gradient forms: high ECM near tumor, declining with distance. The tumor is progressively enclosed. **Failure means:** The paracrine secretion (Knobs 1a, 1b) → stromal activation → ECM production chain is broken. Check TGF-β diffusion, PSC activation threshold, and ACTA2→COL1A1/HAS2 linkage.

Anchor 2: Drug Diffusion Depth Inversely Correlates With Barrier Maturity

Biological fact

PDAC is among the most drug-impermeable solid tumors. As desmoplasia matures, drug penetration from the vasculature (or domain boundary) into the tumor core collapses. Early-stage tumors with thin stroma are more drug-accessible than late-stage tumors with dense stroma.

Simulation test

Setup: Run Anchor 1 to completion (mature barrier). Then introduce a drug substrate at the domain boundary at a fixed concentration (simulating vascular delivery). **Measure:** Drug concentration at the tumor centroid after a fixed diffusion period. Repeat at three time points during barrier formation: early (thin stroma), mid (partial stroma), late (mature stroma). **Expected outcome:** Drug concentration at the tumor centroid should decline monotonically across the three time points. At mature barrier, drug at the centroid should be a small fraction of the boundary concentration — the barrier is working as a diffusion shield. **Failure means:** ECM density is not properly modifying the drug diffusion coefficient. Check the ECM→diffusion coupling in the microenvironment PDE settings.

Anchor 3: SHH Inhibition Alone Accelerates Tumor Growth

Biological fact

The vismodegib Phase II trial in PDAC showed that blocking Sonic Hedgehog signaling — which partially depletes stroma — actually worsened patient outcomes. The stroma is both a barrier to treatment *and* a physical restraint on tumor expansion.

Simulation test

Setup: Run two matched simulations from identical initial conditions to the same endpoint.

- **Control:** No intervention. Barrier forms normally.

- **Intervention:** Set shh_secretion_rate = 0 at simulation start (mimicking vismodegib). No concurrent cytotoxic drug. **Measure:** Total tumor cell count and tumor spatial extent (maximum radius) at endpoint. **Expected outcome:** The intervention arm has *more* tumor cells and a *larger* spatial extent than control. Stroma is partially thinned (GLI1 arm removed, TGF-β arm intact), and the tumor expands into the freed space without any drug to kill the now-accessible cells. **Failure means:** The model is missing the stroma-as-restraint feedback. Either the mechanical confinement (Knob 8) is too weak, or the tumor's proliferative response to reduced compression isn't implemented.

Anchor 4: TGF-β Is Tumor-Suppressive When SMAD4 Is Intact

Biological fact

In the ~45% of PDAC cases where SMAD4 is retained, TGF-β still functions as a growth brake on the tumor. SMAD4-intact PDAC has a different (somewhat better) prognosis and a qualitatively different stroma relationship.

Simulation test

Setup: Run two matched simulations.

- **SMAD4-null:** Default PDAC genotype (tgfb_brake_sensitivity = 0.05). Canonical stroma world.
- **SMAD4-intact:** Same genotype but tgfb_brake_sensitivity = 0.8. TGF-β can now arrest the tumor. **Measure:** Tumor cell count, proliferation rate, and barrier density at endpoint. **Expected outcome:** SMAD4-intact tumors grow slower — TGF-β from the stroma brakes the tumor that built it. Paradoxically, the barrier may be *thinner* because slower tumor growth creates less hypoxic pressure and less paracrine drive. The tumor is partially self-regulating through its own stroma. **Failure means:** The SMAD4 asymmetry (Knob 2, Veto Rule 3 from the Gene State doc) is not properly wired. TGF-β is behaving the same regardless of SMAD4 status.

Anchor 5: Hypoxia Develops in the Tumor Interior as the Barrier Matures

Biological fact

PDAC tumors are profoundly hypoxic — among the most hypoxic of all solid tumors. This hypoxia is largely *caused* by the stroma: dense ECM compresses vasculature and increases diffusion distances. Hypoxia develops from the center outward as the barrier thickens.

Simulation test

Setup: Run Anchor 1 (barrier self-assembly) with oxygen as a tracked microenvironment substrate, supplied from the domain boundary. **Measure:** Oxygen concentration profile across the tumor radius at early, mid, and late time points. **Expected outcome:** Early — relatively uniform oxygenation. Mid — mild gradient begins, center slightly lower. Late — steep gradient, central tumor severely hypoxic, periphery moderately oxygenated. HIF1A should be active in central cells and inactive in peripheral cells. The hypoxic zone should roughly correspond to the region enclosed by the densest ECM. **Failure means:** Either oxygen diffusion is not being impeded by ECM (check ECM→O_2 diffusion coupling), or the domain is too small / boundary O_2 too high for a meaningful gradient to form.

Anchor 6: EMT Is Spatially Restricted to the Tumor-Stroma Interface

Biological fact

EMT in PDAC is not a whole-tumor event. It occurs preferentially at the invasive front — where tumor cells contact stroma, experience TGF-β exposure, and sit in hypoxic/mechanically stressed zones. The tumor core remains more epithelial.

Simulation test

Setup: Run Anchor 1 to mature barrier. At endpoint, query the EMT status (ZEB1 activity, CDH1 expression) of every tumor cell. **Measure:** Fraction of cells in mesenchymal state as a function of distance from the tumor centroid. **Expected outcome:** Mesenchymal fraction should increase with

distance from center — low EMT in the core, high EMT at the periphery. A ring or shell of mesenchymal cells should be visible at the tumor-stroma boundary. The transition should be gradual, not a sharp line. **Failure means:** If EMT is uniform throughout the tumor, emt_induction_threshold (Knob 5a) is too low or TGF-β is flooding the entire tumor instead of being concentrated at the interface. If EMT is absent everywhere, the threshold is too high or TGF-β signaling → ZEB1 activation is disconnected.

Anchor 7: Enzymatic ECM Degradation Improves Drug Delivery But Is Insufficient Alone

Biological fact

Hyaluronidase (PEGPH20) trials showed that degrading HA improved drug delivery biomarkers (reduced interstitial pressure) but did not improve overall survival as monotherapy. ECM degradation opens the physical barrier but doesn't kill the tumor behind it, and the tumor rapidly rebuilds.

Simulation test

Setup: Run three matched simulations from mature barrier state.

- **Control:** No intervention.
- **ECM degradation only:** Apply a global ECM degradation factor (simulating hyaluronidase) that reduces ECM density by a large fraction. No drug.
- **ECM degradation + drug:** Same ECM degradation, plus drug delivered from boundary. **Measure:** ECM density profile, drug penetration depth, and tumor cell viability at endpoint. **Expected outcome:**
- ECM degradation only: Barrier thins, drug diffusion *would* improve but no drug is present. Tumor grows faster (physical restraint reduced, same logic as Anchor 3). ECM begins to re-accumulate within a few simulated days as CAFs continue producing.
- ECM degradation + drug: Drug penetrates deeper. Tumor kill is improved over control. But tumor cells with active ABCB1 (Knob 7b) in the periphery survive, and the barrier begins rebuilding from

surviving CAFs. **Failure means:** If ECM doesn't regrow after degradation, the CAF production loop has been disrupted — check that ACTA2-active CAFs are still present and secreting. If drug + degradation shows no improvement over control, the ECM→diffusion coupling is too weak.

Anchor 8: The Stroma Contains Two Mechanistically Distinct Barrier Components

Biological fact

PDAC stroma is not a uniform wall. It has at least two major components with different physical properties: hyaluronic acid (gel-like, creates interstitial pressure, produced via HAS2) and fibrillar collagen (structural scaffold, creates physical rigidity, produced via COL1A1). They respond to different interventions.

Simulation test

Setup: Run three matched simulations from mature barrier state.

- **HA depletion:** Reduce HAS2 output to zero (simulating hyaluronidase). COL1A1 untouched.
- **Collagen depletion:** Reduce COL1A1 output to zero (simulating collagenase). HAS2 untouched.
- **Both depleted:** Both set to zero simultaneously. **Measure:** Total ECM density, drug diffusion depth, interstitial pressure proxy, and tumor spatial expansion for each. **Expected outcome:** HA depletion should primarily reduce interstitial pressure and improve small-molecule diffusion. Collagen depletion should primarily reduce structural rigidity and release mechanical confinement (tumor expands). Neither alone should fully eliminate the barrier. Both together should produce the largest drug penetration improvement — but also the largest tumor expansion, recapitulating the clinical dilemma. **Failure means:** If HA and collagen depletion produce identical effects, they're not mechanistically distinct in the model — check that they map to different physical properties (diffusion coefficient vs. mechanical stiffness). If depleting both shows no synergy beyond either alone, the two components are not contributing to barrier function through different pathways.

Anchor 9: Barrier Density Predicts Drug Failure Before Drug Arrives

Biological fact

Clinical observation: the denser the stroma at diagnosis (measured by αSMA or collagen staining), the worse the prognosis and the poorer the response to standard chemotherapy. Stroma density is prognostic *independent* of tumor biology — the same tumor behind a thicker wall does worse.

Simulation test

Setup: Run a parameter sweep over initial PSC seeding density (low, medium, high) with the same tumor genotype and same drug regimen. This creates tumors with identical genetics but different barrier thickness at the time of treatment. **Measure:** Tumor cell viability after a standard drug exposure period for each PSC density condition. **Expected outcome:** Strict monotonic relationship — higher initial PSC density → denser barrier at treatment time → less drug penetration → more surviving tumor cells. The same drug at the same dose fails progressively as barrier thickens. This relationship should hold even though tumor genotype (including all death resistance knobs) is identical across conditions. **Failure means:** If drug efficacy is independent of barrier density, the ECM→drug diffusion coupling is too weak or the drug is being delivered in a way that bypasses the barrier (check boundary conditions). If the relationship is non-monotonic, there may be an unintended feedback where intermediate stroma densities create a worse outcome than high density (possible but needs investigation).

Anchor 10: The Barrier Creates a Spatial Sanctuary That Protects a Residual Tumor Population

Biological fact

PDAC almost never achieves complete pathological response to chemotherapy. Even aggressive regimens (FOLFIRINOX) leave viable tumor cells — and these survivors are preferentially located in the most stroma-dense regions where drug concentration was lowest. The barrier creates a spatial sanctuary.

Simulation test

Setup: Run Anchor 1 to mature barrier. Apply a strong drug (high concentration at boundary, long exposure) — designed to be potent enough to kill any tumor cell it reaches at sufficient concentration. **Measure:** After drug exposure, map the spatial distribution of surviving tumor cells against the ECM density field. **Expected outcome:** Surviving cells should be spatially concentrated in the regions of highest ECM density — the deepest interior of the barrier where drug concentration was lowest. The tumor periphery (closest to drug source, thinnest barrier) should show the most cell death. A residual population persists in the sanctuary zone, and these cells remain capable of repopulating if drug is withdrawn. **Post-drug test:** Remove drug and continue the simulation. The sanctuary population should begin re-expanding, secreting stromagenic signals, and rebuilding the barrier around itself — recapitulating clinical recurrence. **Failure means:** If surviving cells are randomly distributed rather than concentrated behind dense ECM, either the drug is penetrating uniformly (barrier not functional) or cell death is stochastic rather than concentration-dependent (check drug pharmacodynamics). If the sanctuary population doesn't regrow after drug withdrawal, the surviving cells have lost their paracrine secretion or proliferative capacity (check that surviving cell genotype is intact).

Anchor Summary Card

#	Anchor	Tests	Core dependency
1	Barrier self-assembles	Paracrine loop intact	Knobs 1a, 1b → stroma activation → ECM
2	Drug penetration drops with barrier maturity	ECM blocks diffusion	ECM → diffusion coefficient coupling
3	SHH inhibition alone worsens outcome	Stroma restrains tumor	Knob 1b, mechanical compaction (Knob 8)

#	Anchor	Tests	Core dependency
4	SMAD4-intact TGF-β brakes tumor	SMAD4 asymmetry works	Knob 2, TGF-β dual-nature wiring
5	Hypoxia develops centrally	O_2 blocked by ECM	ECM → O_2 diffusion, domain geometry
6	EMT is peripheral, not global	Spatial signal gradients	Knob 5a, TGF-β/HIF1A → ZEB1 chain
7	ECM degradation helps drug but isn't enough	Barrier rebuilds, tumor adapts	CAF persistence, efflux (Knobs 7a/7b)
8	HA and collagen are distinct barrier arms	Two-component barrier	HAS2 vs COL1A1 → different physics
9	Thicker barrier = worse drug response	Barrier is prognostic	ECM density → drug diffusion, monotonic
10	Survivors cluster in densest stroma	Spatial sanctuary effect	All systems: diffusion + resistance + regrowth

Running Order

These anchors have dependencies. Run them in this order:

```
Anchor 1 (self-assembly)
  └→ Anchor 5 (hypoxia develops)
     └→ Anchor 6 (EMT is spatial)
  └→ Anchor 2 (drug diffusion drops)
     └→ Anchor 9 (density predicts failure)
        └→ Anchor 10 (spatial sanctuary)
  └→ Anchor 3 (SHH paradox)
  └→ Anchor 4 (SMAD4 asymmetry)
  └→ Anchor 7 (ECM degradation insufficient)
     └→ Anchor 8 (two-component barrier)
```

Anchor 1 is the foundation. If the barrier doesn't self-assemble, nothing downstream works. Debug Anchor 1 until it passes before touching anything else.

Anchor 10 is the integration test. It requires every subsystem to be functional simultaneously: paracrine signaling, ECM production, drug diffusion, death resistance, spatial gradients, and regrowth. If Anchors 1–9 pass but Anchor 10 fails, there's an interaction bug that only manifests when all systems run together.

Ten reality anchors. Each one is a fact about PDAC that the model must reproduce or the model is wrong. They are the contract between the simulation and biology.